



free Command in Linux

Introduction

Imagine you are using a **Linux system** and need to:

- Check how much RAM is free and available.
- Find out if your system is running out of memory.
- Monitor swap space usage.

How do you get a quick summary of memory usage?

This is where the **free command** helps!

Section 1: Learn

What is **free**?

The **free** command in Linux displays information about total, used, and available memory, including RAM and swap. It helps:

1. Monitor system performance and prevent slowdowns.
2. Check if more memory (RAM) is required.
3. Optimize memory usage by analyzing free and cached memory.

Why Do We Need **free**?

Without free	With free
No way to check RAM usage	Displays memory usage instantly
System may slow down unexpectedly	Helps detect low memory conditions



Without free	With free
No easy method to check swap usage	Shows swap space usage clearly
Hard to optimize performance	Helps manage resource allocation

How Does **free** Work?

1. You run the **free** command in the terminal.
2. It displays memory details including total, used, and available RAM and swap.
3. You can use options to format and analyze memory statistics.

An Interesting Anecdote

In early computing, systems crashed due to insufficient memory, and users had no way to check RAM usage. The **free** command was introduced in Unix-based systems to give instant memory statistics, helping users troubleshoot performance issues.

Section 2: Practice

Step-by-Step Guide to Using **free**

1. Running the **free** Command

```
# Show memory usage summary  
free
```

What happens here?

- Displays total, used, and free memory (RAM and swap).



2. Displaying Memory in Human-Readable Format (-h)

Show memory in MB/GB instead of bytes

```
free -h
```

Why is this useful?

- Easier to **read memory values in MB and GB** instead of large numbers in bytes.

3. Showing Buffers and Cache (-m)

Show memory details in megabytes

```
free -m
```

Column	Meaning
total	Total RAM installed
used	Memory currently in use
free	Unused memory
shared	Memory used by shared processes
buffers	Temporary data storage in memory
cache	Cached data for faster access
available	Memory available for new applications

How does this help?



- **Cached memory is not wasted memory.** It speeds up system performance!
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4. Displaying Memory Every Few Seconds (**watch**)

```
# Update memory usage every 2 seconds  
watch -n 2 free -h
```

What does this do?

- Continuously updates memory usage, **helpful for real-time monitoring.**
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5. Checking Swap Memory Usage

```
# View swap space usage  
free -h | grep Swap
```

Why is this important?

- Swap acts as **backup RAM**, but excessive swap usage **slows down performance.**
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Real-World Example

A **system administrator** notices **server slowdowns**. They need to:

1. **Check if RAM usage is high.**
2. **See how much swap is being used.**
3. **Decide if more RAM is needed.**

What did they do?



1. Ran **free -h** to analyze memory usage.
2. Used **watch -n 2 free -h** to monitor memory changes in real-time.
3. Found that **swap usage was high**, meaning **RAM was insufficient**.

Result? They **upgraded the RAM and optimized cache settings**, improving server performance!

Section 3: Know More

Frequently Asked Questions

1. How do I check only free memory?

```
free -h | grep Mem
```

2. What is the difference between **free** and **top**?
 - **free** gives a quick memory summary.
 - **top** shows detailed process-wise memory usage.
3. Why is my available memory different from free memory?
 - **Available memory** includes cache that can be freed, while **free memory** is completely unused RAM.
4. How do I check swap usage only?

```
free -m | grep Swap
```

5. Where can I practice memory monitoring?
 - Use a **Linux Virtual Machine**.
 - Monitor memory on **cloud servers like AWS**.
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Conclusion

The **free** command **helps monitor system memory, prevent slowdowns, and optimize performance.**

Start by **checking memory statistics, using human-readable formats, and monitoring swap usage,** and soon you'll be **managing system memory like an expert!**